BME577: Drug Delivery **Spring 2025**
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Project 2: Mathematical Modeling of Systemic Delivery of IgG

1) *Re-write Equation 2 in the cylindrical coordinate system and the expression of Q based on what you learned in class.*

2) *List governing equations for C_B and C_{AB2} based on what you learned in class.*

3) *Convert the Equation 2 (PDE) into ordinary differential equations by replacing the partial derivatives with respect to r in the PDE with finite difference expressions.*

4) *Solving the ODEs and the governing equations for C_B and C_{AB2} simultaneously to obtain C_A and C_{AB2} as functions of time (from 0 to t_{end}) and r (between a and R), and using a 3-D plot tool in python to show results of C_A, C_B, and C_{AB2} times t_{end} = 3 days.*

Figure 1. 3D surface plot depicting the free IgG concentration C_A as a function of radial distance rr (from the microvessel wall at r = a to r = R) and time (0 to 3 days). The solution is obtained by numerically solving the governing PDE with boundary flux at r = a and no flux at r = R, and includes diffusion and reversible binding to antigen.

In this plot, the IgG enters through the microvessel wall at r = a and diffuses outward while undergoing reversible binding. Early on, C_A rises rapidly near the boundary where the antibody is

supplied by the plasma, then spreads radially toward r = R. Over time, as binding and transport continue, the overall concentration diminishes but remains highest closer to the vessel wall, reflecting both the supply of IgG and its gradual consumption through antigen binding.

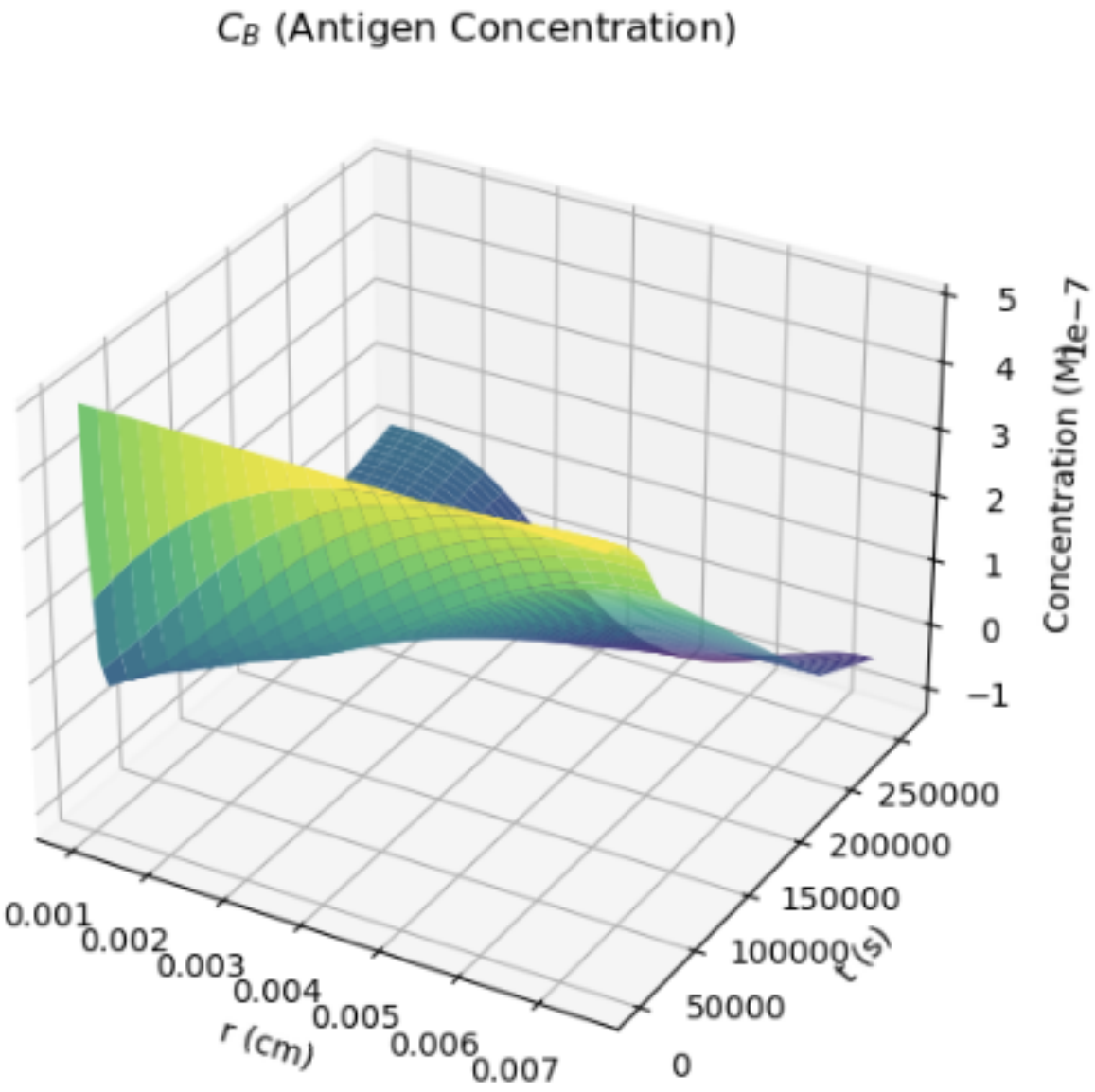


Figure 2. 3D surface plot of the antigen concentration C_B over radial distance r and time. Initially uniform at C_{B0}, the antigen is gradually consumed as it binds free IgG near the vessel wall, leading to a decreasing and spatially varying profile over the 3‑day simulation.

In this plot, the antigen concentration C_B begins at a uniform level throughout the tissue and is progressively depleted where it binds IgG. Over time, the spatial profile reveals that C_B is

consumed most strongly near the microvessel wall, where the antibody flux is highest, leading to a gradient that extends radially outward.

Figure 3. 3D surface plot of the bound IgG–antigen complex C_{AB2} over radial distance r and time. The complex forms as free IgG binds locally to antigen, reaching higher levels where both substrate and antibody concentrations coincide, and accumulates over the 3‑day simulation.

In this plot, the complex C_{AB2} forms wherever free IgG and antigen overlap, causing the concentration to rise from zero as binding proceeds. The highest levels appear in regions with

ample IgG supply (near the vessel) and sufficient remaining antigen, and this bound complex gradually accumulates over the 3‑day simulation period.

5) *Repeat the plot in Assignment 4 if there is no binding between IgG and antigen.*

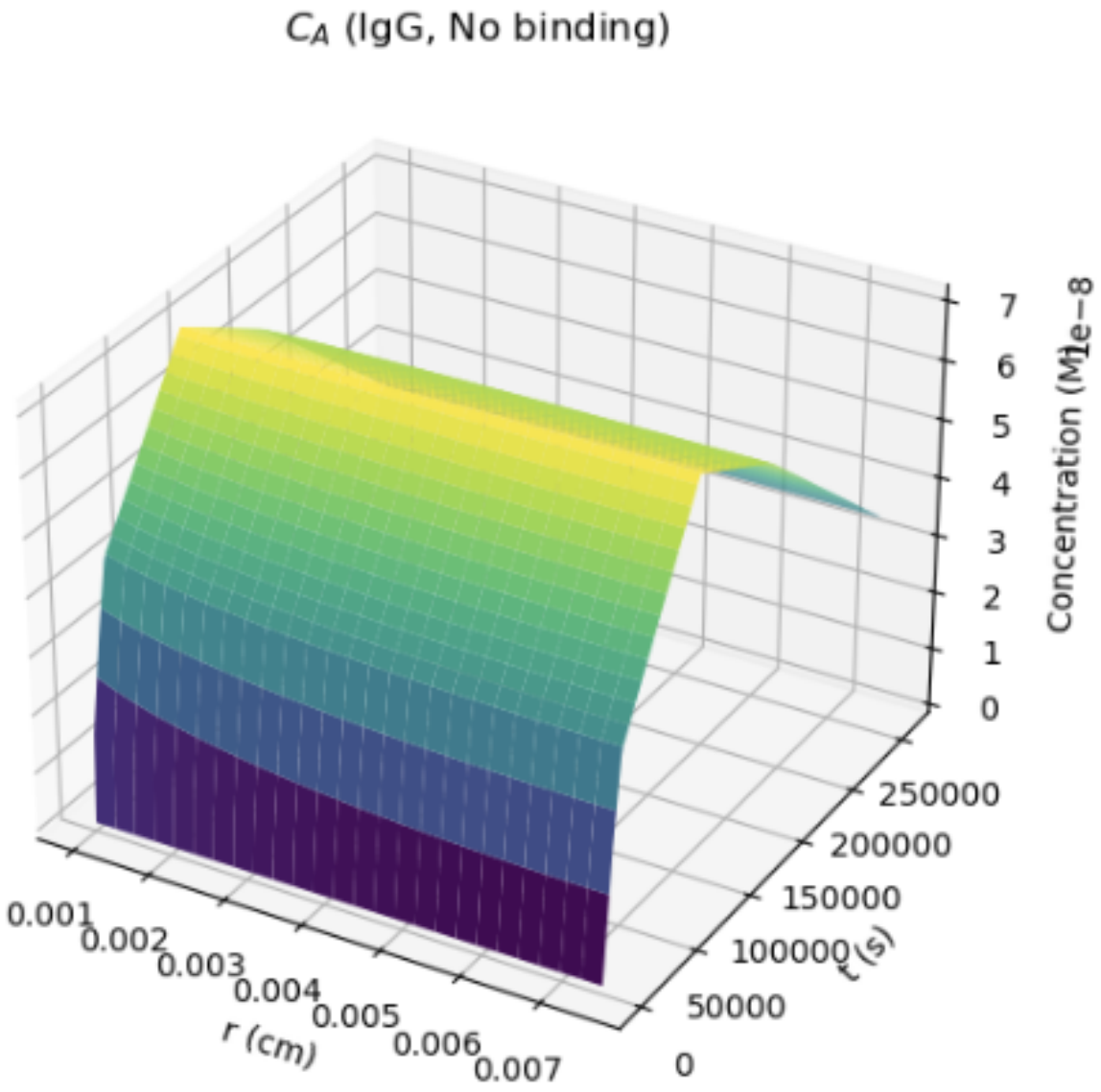


Figure 4. 3D surface plot of the free IgG concentration C_A under no‑binding conditions. IgG still enters the tissue at r = a and diffuses outward to r = R, but in the absence of antigen binding, the concentration profile remains higher and undergoes a purely diffusive spread over the 3‑day simulation.

In this *no-binding* scenario, the IgG spreads purely by diffusion from the microvessel wall, where the concentration is highest, toward the outer boundary at r = R. Since there is no antigen consumption, the overall C_A values remain higher and develop a gentler radial gradient compared to the binding case, reflecting a simple diffusive accumulation over the 3-day period.

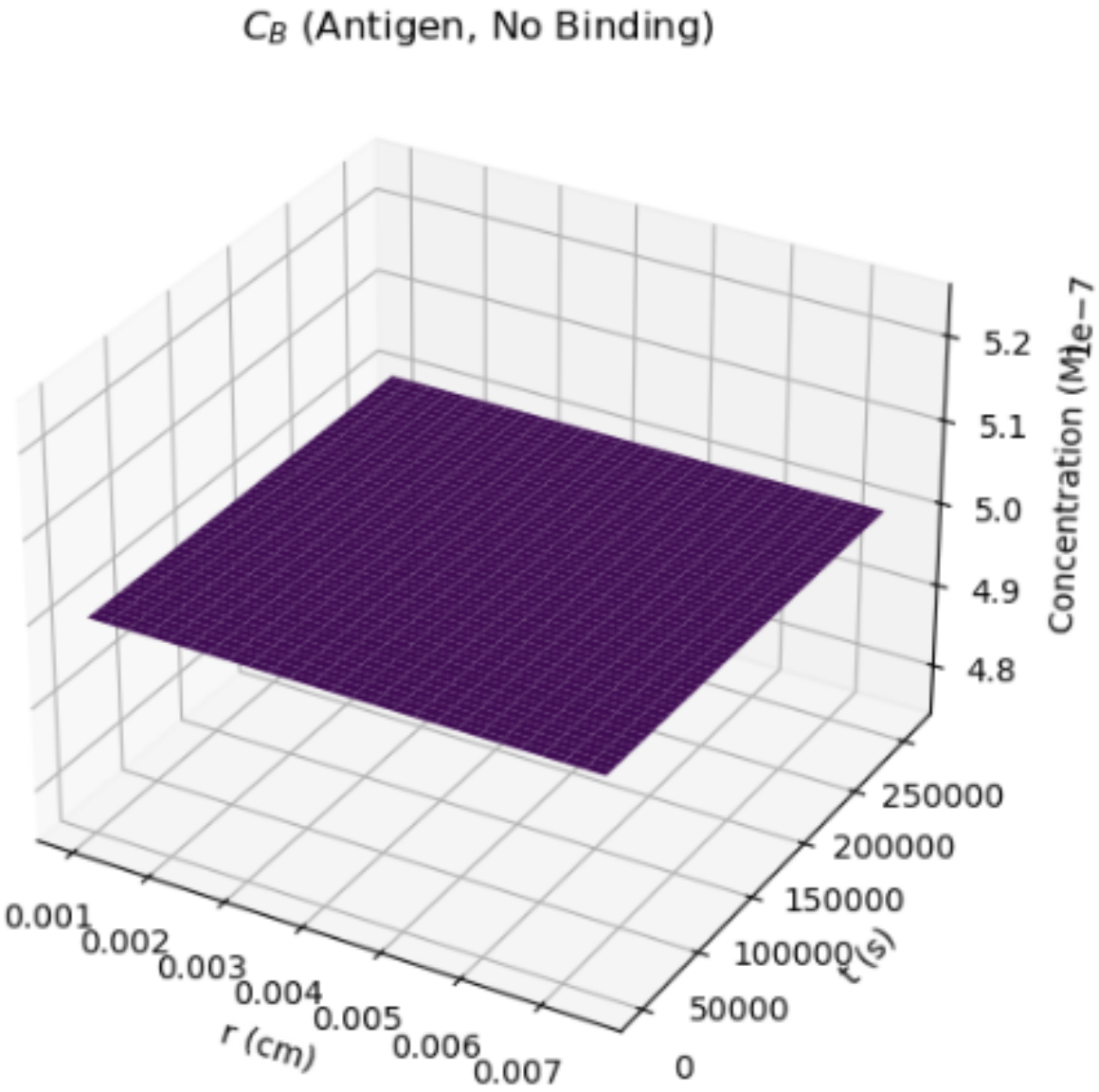


Figure 5. 3D surface plot of the antigen concentration C_B under no-binding conditions. Without any reaction to consume C_B, its profile remains uniform at a concentration of 0.5 μM across the tissue and constant over the 3-day simulation period.

In the *no-binding* scenario, there is no reaction term to consume or produce antigen; consequently, whatever initial antigen concentration is specified remains uniformly constant throughout the tissue and entire simulation. In this example, C_B is initialized at 0.5 μM and, with no mechanism to reduce it, it stays at that level for all radii and all times, resulting in a flat surface in the 3D plot.

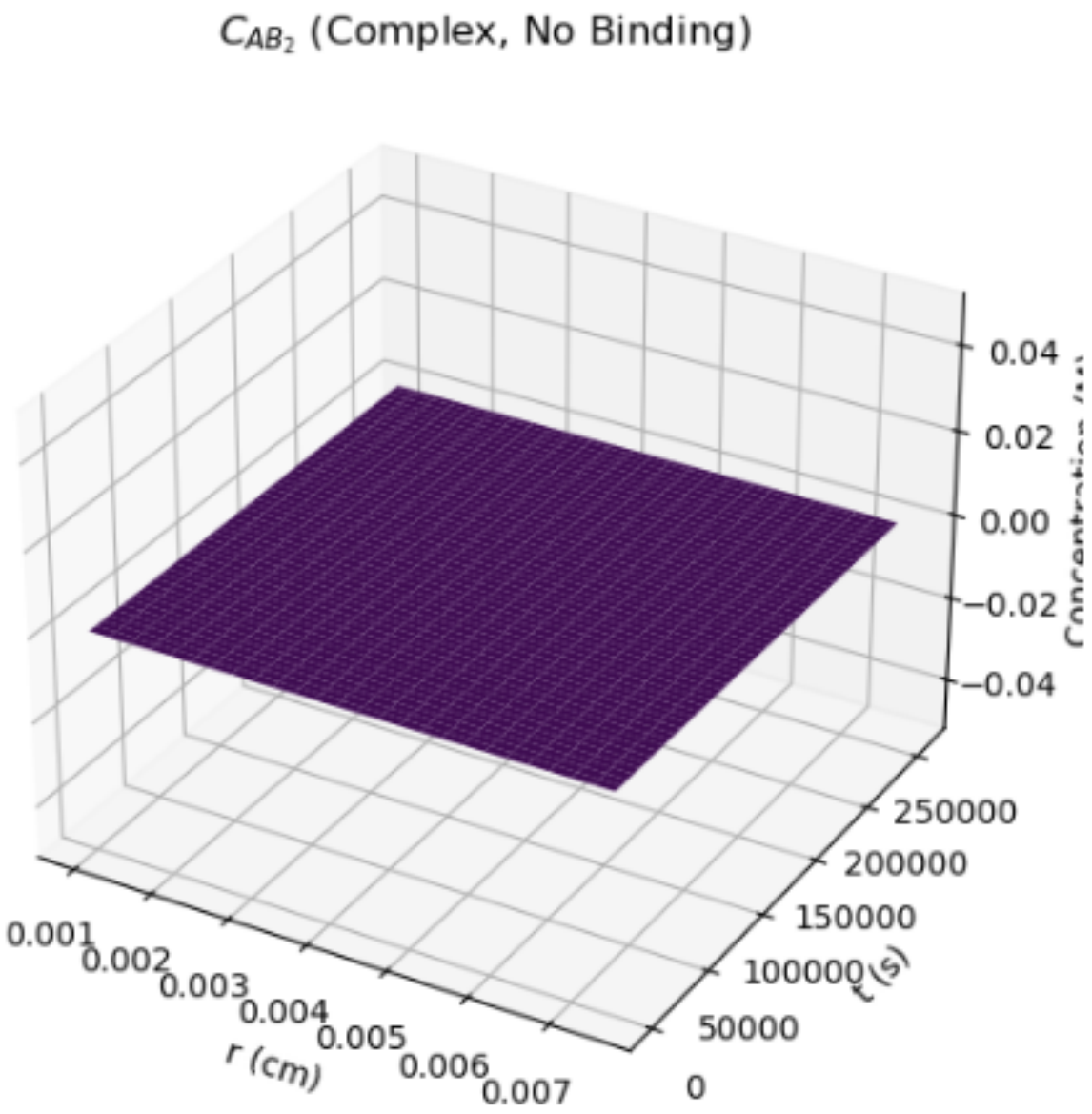


Figure 6. 3D surface plot of the bound complex C_{AB2} under no‑binding conditions. Because no reaction occurs in this scenario, the complex remains at zero for all radial positions and times, producing a uniform, flat surface.

Since there is no reaction to form the antibody–antigen complex, C_{AB2} remains at 0 everywhere in the tissue. The 3D surface is therefore completely flat, indicating that no bound complexes are formed when there is no binding mechanism.

6) *Specificity ratio (SR) is a parameter used to quantify the extent to which antibody/antigen binding enhances antibody delivery to tumors. It is defined as:*

$$SR = C_{specific} / C_{non\text{-}specific}$$

where $C_{specific}$ and $C_{non\text{-}specific}$ are the total concentrations of IgG (= free + bound). Plot SR as a function of time (from 0 to t_{end}) and r (between a and R), using the 3D-plot tool.

Figure 7. 3D surface plot of the specificity ratio $SR = C_{specific} / C_{non\text{-}specific}$ over radial distance r and time. Larger values indicate a greater advantage due to antigen-specific binding (accumulation of IgG) relative to the purely diffusive, no-binding scenario.

In this final 3D plot, the specificity ratio compares the total IgG buildup with antigen binding to the case without binding. A higher SR value indicates that binding is concentrating more antibody within the tissue than diffusion alone could achieve, particularly near the microvessel wall and at later times when a larger fraction of IgG has had time to bind.